



SNG BUILDER SERIES — GUIDE #004 OF 4

E-520 JACKSHAFT FABRICATION & ASSEMBLY

DUAL-STAGE 15.44:1 REDUCTION SYSTEM — NO MACHINIST SKILLS REQUIRED

15.44:1 TOTAL REDUCTION	2-Stage DRIVE SYSTEM	16 Parts EXPLODED BOM	6 Steps ASSEMBLY SEQUENCE	\$49 GUIDE PRICE
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You don't need to be a machinist. Hand the instructions in this guide to any machine shop and they will have zero follow-up questions. Every dimension, every tolerance, every material spec is documented. The plate and jackshaft shaft are the only custom parts — everything else ships off a shelf.

WHAT'S INSIDE THIS GUIDE

- System Overview**
Dual-stage reduction concept, why 15.44:1, where this system fits in the SNG builds
- Machinist Instructions**
Ready-to-hand-over spec sheet for plate and shaft fabrication. Machinist needs nothing else.
- 16-Part Exploded BOM**
Every component numbered, sourced, and priced. What you buy vs. what gets machined.
- Critical Dimensions**
Shaft diameter, bearing spec, plate dimensions, bearing centers — all documented
- 6-Step Assembly Sequence**
Step-by-step with photos. Critical: secondary (rear) chain aligned first — always.
- Torque Specs**
Every fastener torqued to spec. M8, M10, and set screws — no guessing.
- Chain Alignment Procedure**
The sequence that matters. Why rear chain goes first. How to set both stages correctly.
- Installation Notes**
Integration into the full build — connects with Guide #001, #002, and #003 builds

BUILDER'S NOTE — BRYAN JAWORSKI, SNG CYCLES

I'm not a machinist either. That's why I wrote these instructions the way I did — so I could hand them to a machine shop and not get a single call back with questions. The plate and shaft are the only parts that need machining. Everything else — bearings, sprockets, chain, fasteners — comes off a shelf. This guide handles all of it.



REDUCTION SYSTEM — DUAL STAGE

E-520 JACKSHAFT — DRIVE RATIO BREAKDOWN		
Primary Stage	12T Drive Sprocket → 48T Driven Sprocket	4.00:1
Secondary Stage	14T Drive Sprocket → 54T Driven Sprocket	3.86:1
Total Reduction	4.00 × 3.86 — Dual-stage compound reduction	15.44:1

CRITICAL DIMENSIONS

20mm JACKSHAFT DIAMETER	UCP204 BEARING TYPE	197mm BEARING CENTER-TO-CENTER
3/8" PLATE THICKNESS	A36/1018 PLATE MATERIAL (MILD STEEL)	#420 CHAIN TYPE

TORQUE SPECIFICATIONS

22 Nm M8 FASTENERS	40-45 Nm M10 FASTENERS	10-12 Nm SET SCREWS
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WHAT NEEDS MACHINING VS. WHAT YOU BUY

THE KEY POINT

Only two parts need machining: the mounting plate and the jackshaft. Everything else — bearings, sprockets, chain, fasteners, hardware — is off-the-shelf. The guide includes the exact machinist instructions for both custom parts. Hand them to any machine shop. No follow-up questions. No back-and-forth.

COMPONENT	HOW TO GET IT
Mounting Plate — 3/8" Mild Steel A36/1018	Machine shop — instructions in guide. Exact dimensions, hole pattern, and spec sheet ready to hand over.
Jackshaft — 20mm	Machine shop — instructions in guide. Machinist gets the drawing and asks no further questions.
UCP204 Pillow Block Bearings	Off the shelf — eBay / bearing supplier
Primary Sprockets (12T + 48T)	Off the shelf — eBay / sprocket supplier
Secondary Sprockets (14T + 54T)	Off the shelf — eBay / sprocket supplier
#420 Chain (Primary)	Off the shelf — length in guide
Chain (Secondary/Final Drive)	Off the shelf — per guide spec
Fastener Kit (M8 + M10 + set screws)	Off the shelf — hardware store or eBay

6-STEP ASSEMBLY SEQUENCE

- 1

RECEIVE MACHINED PARTS — INSPECT BEFORE ASSEMBLY

Verify plate dimensions and shaft diameter before doing anything else. Check bearing bore fit on shaft. Full inspection checklist in guide.
- 2

MOUNT UCP204 BEARINGS TO PLATE

197mm center-to-center. Torque bearing mount hardware to spec. Do not fully tighten until shaft is installed and both chains are set.

⚠ Do not fully tighten bearing mounts at this stage.
- 3

INSTALL JACKSHAFT THROUGH BEARINGS

Slide 20mm shaft through both UCP204 bearing bores. Stage your sprockets on the shaft in correct position before proceeding — position spec in guide.
- 4

SET SECONDARY (REAR) CHAIN FIRST — ALWAYS

Align the 14T→54T secondary stage chain before touching the primary. This is the critical step. Chain alignment sequence is documented in detail in the guide.

⚠ Critical: Secondary chain alignment first — every time. Primary chain alignment depends on this position.
- 5

SET PRIMARY CHAIN — THEN LOCK EVERYTHING

With secondary set, align primary 12T→48T chain. When both chains are running true with correct tension, fully torque all bearing mounts and set screws to spec.

M8: 22 Nm · M10: 40-45 Nm · Set Screws: 10-12 Nm
- 6

FINAL INSPECTION & INSTALLATION

Rotation check, chain tension verify, torque confirmation. Mount to frame per build guide (compatible with Guide #001, #002, and #003 builds). Full installation notes in guide.



ADDITIONAL NOTES — WP SUSPENSION & BRAKES

WP FORK CONVERSION — LEWIS BRAKES ADAPTER

When the SNG builds were in development, no ready-made adapter existed for WP fork conversion with the Lewis Brakes system. SNG developed CAD drawings for the adapter at that time. Since then, Lewis Brakes has released a complete kit at \$249 — and this is now the recommended path. Even if you fabricated the adapter yourself, you'd still need a special notched axle and different spacer. The math doesn't work for DIY. Buy the Lewis Brakes kit.

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GET THE FULL GUIDE

This document is the **product overview and technical index**. The complete Guide #004 includes the machinist-ready fabrication spec sheet, full exploded BOM with part numbers, assembly photos for all 6 steps, and installation notes for each SNG build platform.

\$49

GUIDE #004 —
JACKSHAFT

\$149

GUIDE #001 — LBX
FRAME-UP

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GUIDE #002 — 270-E
BUILD

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